

Bonus 5 - Besten!

$$\textcircled{1} \int \left(2 - 3x^3 + \frac{1}{x^2} \right) dx = 2x - 3 \cdot \frac{x^4}{4} + \frac{x^{-1}}{-1} + C = \underline{2x - \frac{3x^4}{4} - \frac{1}{x} + C}$$

$$\textcircled{2} \int x \cdot \cos(x^2+5) dx = \left| \begin{array}{l} t = x^2+5 \\ dt = 2x dx \end{array} \right| = \frac{1}{2} \int \cos t dt = \underline{\frac{1}{2} \sin(x^2+5) + C}$$

$$\textcircled{3} \int (5x+2)^5 dx = \frac{1}{5} \cdot \frac{(5x+2)^6}{6} + C = \underline{\underline{\frac{(5x+2)^6}{30} + C}}$$

$$\textcircled{4} \int_1^2 (x^2+2x) dx = \left[\frac{x^3}{3} + x^2 \right]_1^2 = \frac{8}{3} + 4 - \frac{1}{3} - 1 = \frac{7}{3} + 3 = \underline{\underline{\frac{16}{3}}}$$